ANIKET KULKARNI

VLSI Design & Technology | RTL Design | Verification | Physical Design

Pune, Maharashtra | Expected graduation: 2027

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PROFESSIONAL SUMMARY

Final-year VLSI Design & Technology undergraduate with hands-on work in processor RTL, arithmetic datapaths, memory blocks, directed verification, and physical-design analysis. Built a five-stage RV32IM + Zicsr microcontroller SoC and published seven documented Verilog projects with source, testbenches, and verification records.

TECHNICAL SKILLS

RTL & verification: Verilog, SystemVerilog, Icarus Verilog, VCS/Verdi, directed testbenches, waveform debugging

Architecture: RISC-V RV32IM + Zicsr, MIPS-style CPUs, pipelining, FIFOs, fixed-point Q7.8, CSR/traps, SPI

Implementation: Yosys, OpenLane/SKY130, OpenROAD/OpenSTA, KLayout, Magic/Netgen; Design Compiler and IC Compiler II report analysis

SELECTED VLSI PROJECTS

32-Bit RISC-V Microcontroller SoC | RV32IM + Zicsr | OpenLane / SKY130 | Oct 2026

- Designed a five-stage CPU with forwarding/hazards, a 32-cycle multiply/divide unit, machine-mode traps, UART, GPIO, timer, SPI, and synchronous WFI sleep control; directed firmware simulation passed.
- Saved phase-5 OpenLane reports record a completed 1 mm² layout with 21,584 routed cells and zero Magic DRC and Netgen LVS errors; extracted setup timing and antenna checks remain open.
- OpenSTA typical-corner switching estimate fell 28.2% versus phase 4 (9.65 to 6.93 mW) after sleep-control and physical-design changes.

32-Bit Five-Stage MIPS-Subset Processor | Verilog RTL | 2026

- Implemented forwarding, load-use stalls, branch/jump flushing, and instruction/data memories; directed tests checked six registers and one memory result; Yosys generic synthesis passed.

4 × 4 Fixed-Point Matrix Multiplier | Verilog RTL | 2026

- Built a sign-magnitude Q7.8 datapath with 64 multipliers and 48 adders; the supplied testbench checked all 16 outputs for one matrix pair; Yosys generic synthesis passed.

16 × 32 Synchronous FIFO | Verilog RTL | 2026

- Published parameterized and structural single-clock implementations; eight directed suites exercise data movement, reset, full/empty, threshold, and boundary behavior.

Additional RTL Projects | 16-Bit MIPS-Style CPU, 8-Bit Datapath & 4-Bit ALU | 2026

- Published source and directed simulations for a custom-ISA processor, externally instructed ALU/register datapath, and registered eight-operation ALU.

RELEVANT EXPERIENCE

Digital Electronics & VLSI Intern | Codec Technologies Pvt. Ltd. | Jul–Aug 2026

- Completed structured training in digital logic, number systems/codes, and core VLSI concepts during a two-month internship.

Technical Assistant Intern | Dynomerk Controls India Pvt. Ltd. | Jan–Feb 2026

- Assisted with industrial control-panel assembly and electronic components; used AutoCAD and automation testing tools for Engine and ABS systems.

EDUCATION

Undergraduate Degree in VLSI Design & Technology | APCOER, Pune | 2023–2027

Anantrao Pawar College of Engineering and Research | Savitribai Phule Pune University

WORKSHOPS, CERTIFICATIONS & TRAINING

Semiconductors for AI & AI for Semiconductors | SemiX AI SIG, IIT Bombay | 17 Oct 2026

Guided reference-design walkthrough: 4 × 4 systolic array (16 PEs), skewed dataflow and golden-model checks; Synopsys dc_shell / ICC2 flow from synthesis through routing.

- 4th Annual SemiX Summit, IIT Bombay, 14–17 Oct 2026: semiconductor entrepreneurship, AI compute, and power electronics.
- Physical Design Using Synopsys — VLSIGuru, Sep 2026; VLSI Specialist Skillset Test — Beep, Aug 2026.
- Semiconductor Manufacturing — CeNSE, IISc, Apr 2026; RTL to GDSII — SemiconTantra, Feb 2026.
- Verilog HDL Fundamentals — Udemy, Oct 2025; AI PC Workshop — SemiX, IIT Bombay, Sep 2025.

LEADERSHIP

Technical Symposium Coordinator, SEMICONIX – APCOER: co-organized departmental quizzes and presentations.